DEEPESH BHATIYA

GenAI / Backend Engineer

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PROFESSIONAL SUMMARY

GenAI Engineer with 1.2 years of experience designing and building multi-agent, LLM-powered backend systems for real-world business workflows. Skilled in Python, LangChain/LangGraph, CrewAI, FastAPI, and vector databases, with a strong focus on agentic application development, orchestration, and retrieval-augmented generation (RAG). Comfortable owning systems end-to-end, from agent design to API development and deployment, with a primary interest in GenAI application and backend engineering .

WORK EXPERIENCE

GenAI Engineer - Fractal Analytics *Aug 2025 - Oct 2026*

Project: TradeOps

- Built TradeOps, an agentic trade-settlement automation platform using FastAPI, SQLite, LangGraph, and GPT-4o-mini to monitor trade breaks, classify issues, and automate broker and middle-office follow-up communication via Microsoft Graph.
- Designed a multi-agent workflow with rule-based orchestration coordinating 8 agents simultaneously on live data refreshed every 5 minutes, covering counterparty analysis, context-aware email drafting, and reply handling.
- Implemented automated reminders and escalations for unresolved trade breaks and automated tests for the agent pipeline, streamlining settlement operations end-to-end from issue detection to resolution.
- Architected scalable backend systems for analytics and chatbot applications, applying multi-source conversational AI and agentic workflows (Langgraph) to enterprise document Q&A use cases.

PROJECTS

NL-to-SQL Query Engine with Auto-Visualization *Tech: Python, FastAPI, LangChain, Llama 3.1, Azure SQL*

- Designed and built an end-to-end Text-to-SQL platform using Python, FastAPI, LangChain, Llama 3.1, and Azure SQL, enabling secure natural-language-to-SQL query generation and execution for non-technical users.
- Built an LLM-driven autonomous pipeline for intent detection, SQL validation, and conversational query handling, with automated Azure-based data ingestion from CSV/Excel and schema and metadata management.
- Added a results-visualization layer that profiles query output and auto-selects the most suitable chart type, turning raw query results into instant, presentable insights.

PDF RAG Chatbot with RAGAS-Based Evaluation *Tech: Python, LangChain (LCEL), FAISS, Groq Llama 3.1, RAGAS*

- Developed a RAG-based chatbot for PDF document Q&A using FAISS, LangChain (LCEL), and Groq Llama 3.1, with retrieval reranking to improve answer accuracy and relevance.
- Integrated the RAGAS evaluation framework to continuously benchmark faithfulness, answer relevancy, and context precision, enabling data-driven pipeline improvements.

EDUCATION

National Institute of Technology (NIT), Jamshedpur *2022 - 2025*
Master of Computer Applications (MCA) *CGPA: 7.99*

Shri Vaishnav Institute of Management, Indore *2018 - 2021*
Bachelor of Computer Applications (BCA) *71.71%*

TECHNICAL SKILLS

Languages: Python, C++, JavaScript

GenAI / Agentic Frameworks: LangChain, LangGraph, CrewAI, Agentic AI, RAG

Backend & APIs: FastAPI

Databases: PostgreSQL, MongoDB, SQL

Vector Databases: FAISS, ChromaDB

Cloud & Tools: Azure (basics), Git, GitHub, Postman, VS Code

CS Fundamentals: Data Structures & Algorithms, Operating Systems, Computer Networks, DBMS, OOP

TECHNICAL ACHIEVEMENTS

- LeetCode - Highest rating 1622; 500+ problems solved across varied patterns.
- CodeChef - Highest rating 1619; 200+ problems solved; Global Ranks 188 & 275 (Starter), 130 & 145 (Div 3).
- Joint Coordinator for a fun/cultural event at NIT Jamshedpur; completed a 7-day NSS volunteer camp focused on teamwork and social responsibility.